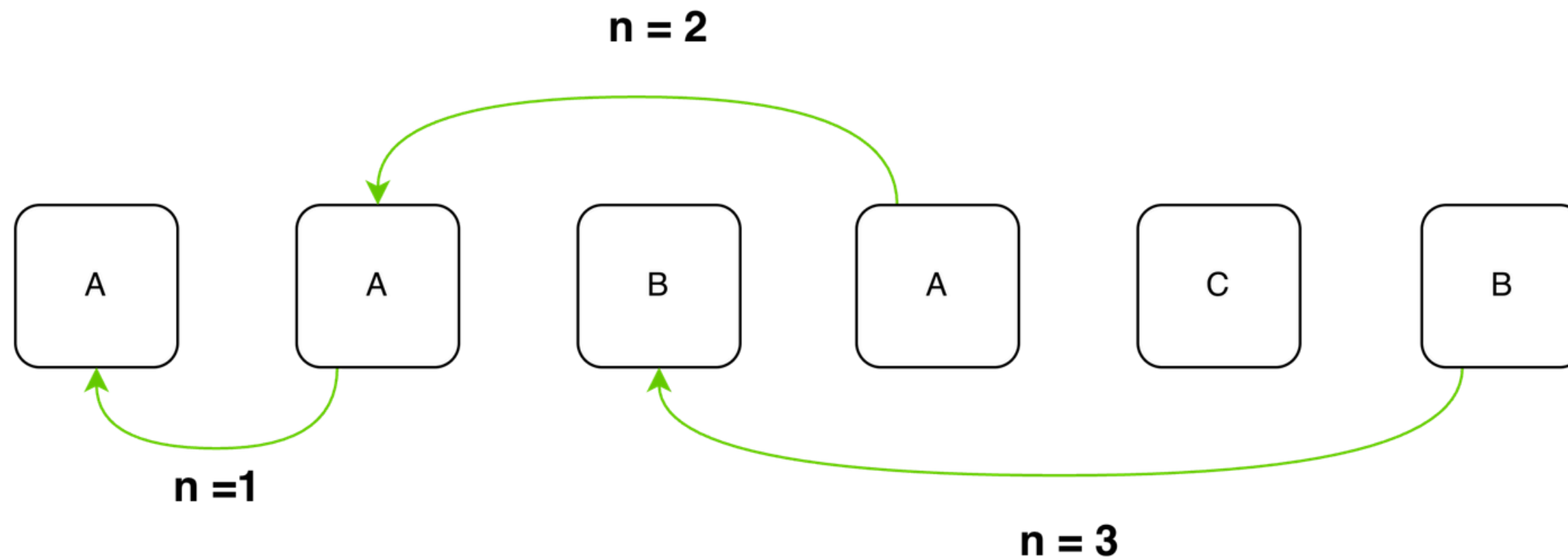


FTEC 5660 Reproducibility Project

**WORKING MEMORY CAPACITY
OF CHATGPT:
AN EMPIRICAL STUDY**

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1 March 2026

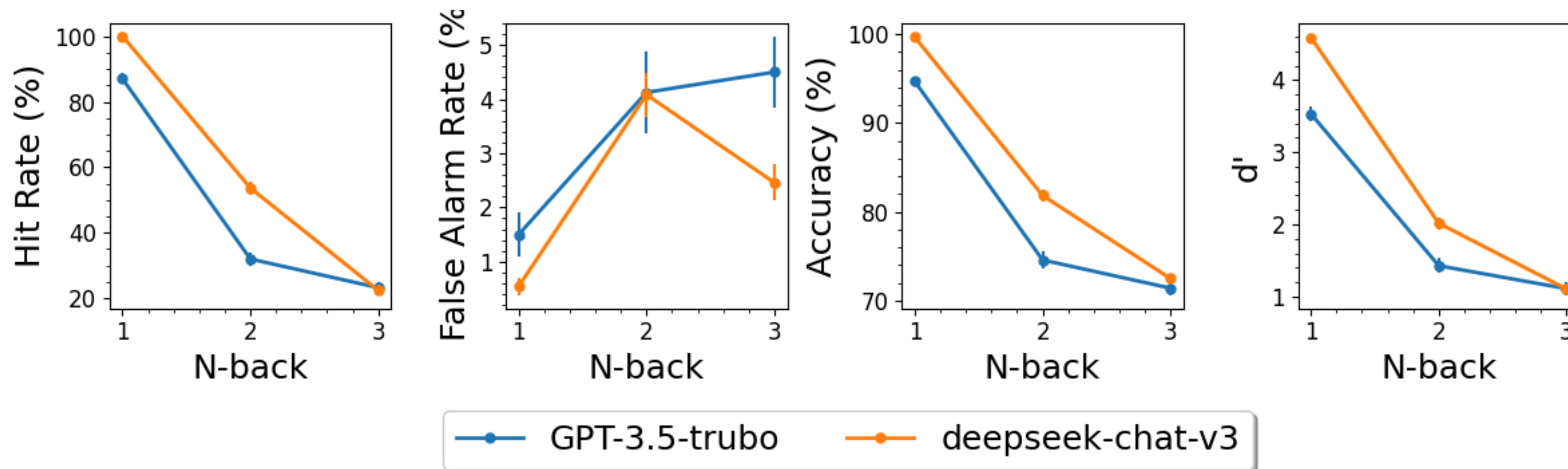
BACKGROUND: THE N-BACK TASK



REPRODUCTION SCOPE & CONSTRAINTS

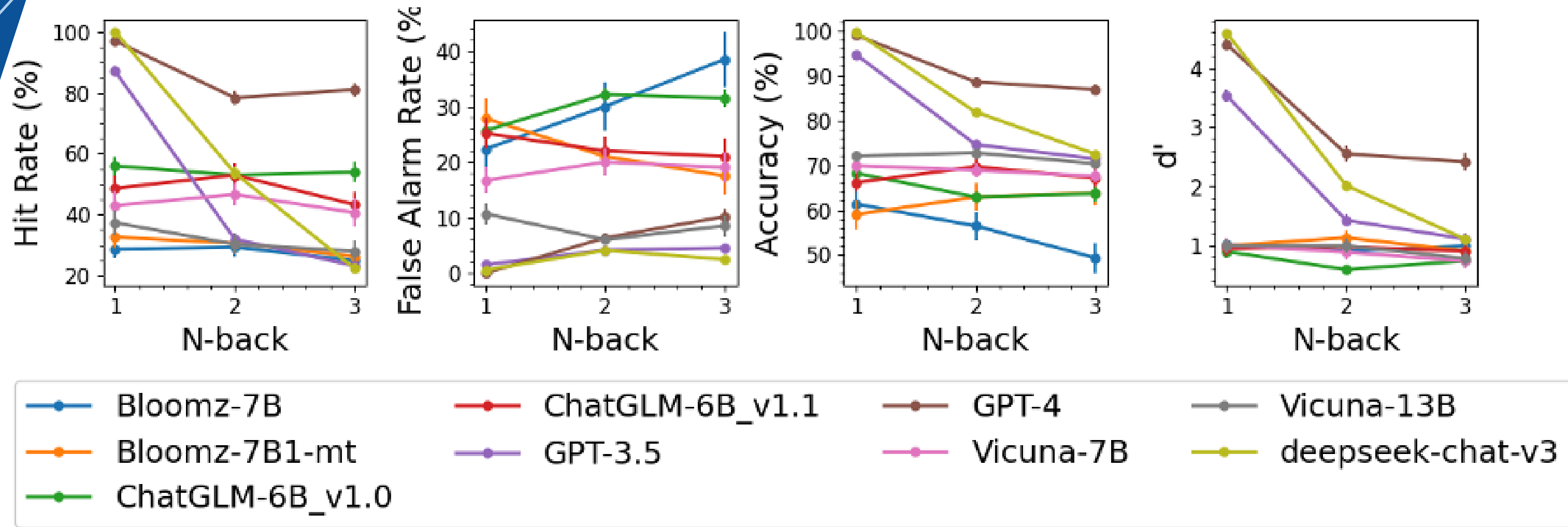
- **Methodology:** Verbal Base Task (Letters).
- **Model Substitution:** GPT-3.5-Turbo (Original) -> DeepSeek-V3.2 (Reproduced).
- **Scale:** 150 total blocks (3 Runs of 50 Blocks).
- **Primary Metric:** d' (Detection Sensitivity), calculated as $d' = Z(\text{Hit Rate}) - Z(\text{False Alarm Rate})$.
- **Other Metrics:** Hit Rate, False Alarm Rate, Accuracy

RESULTS: ORIGINAL VS REPRODUCED



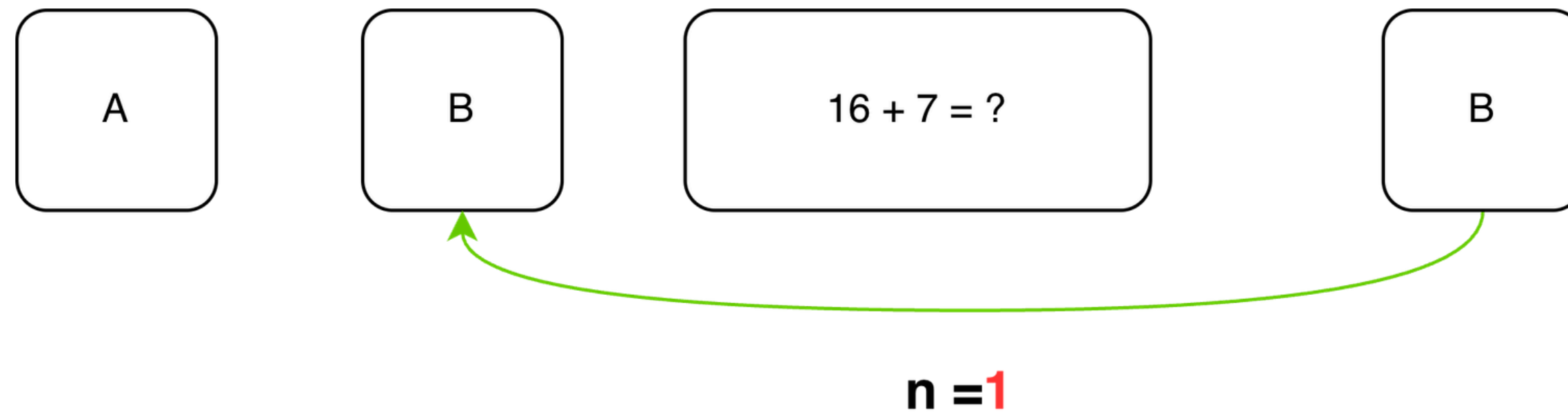
Observation: DeepSeek outperforms GPT-3.5 at n=1, 2 but crashes at n=3

RESULTS: CROSS-MODEL COMPARISON



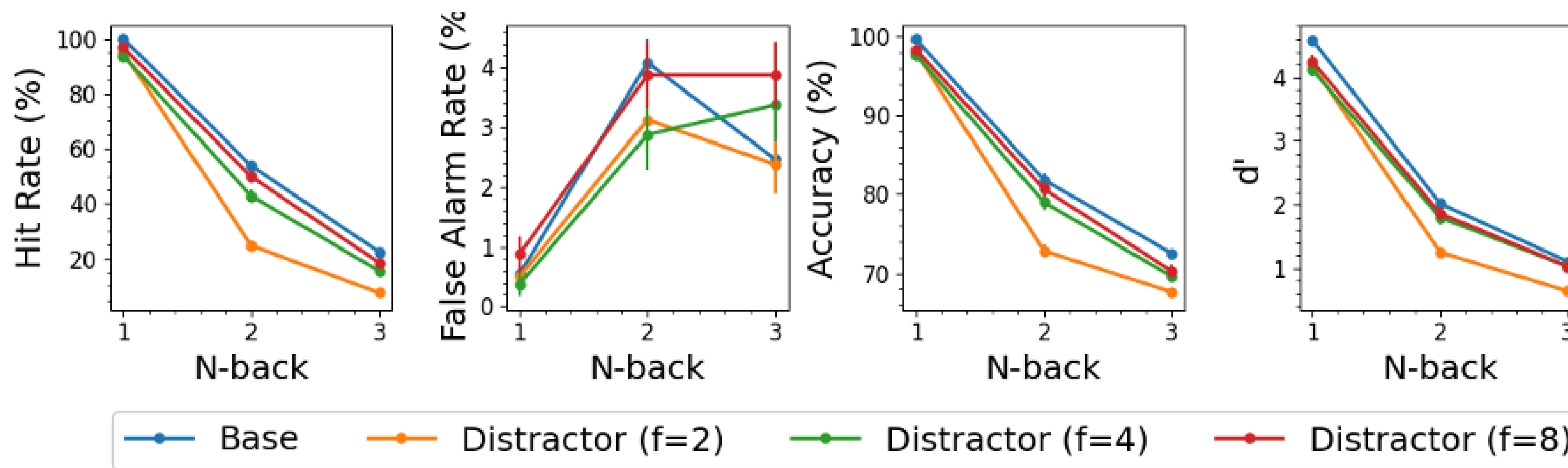
- **n=1:** DeepSeek ~ GPT-4.
- **n=2:** DeepSeek > Most models, but < GPT-4.
- **n=3:** DeepSeek ~ Most models (Convergence), < GPT4

MAJOR MODIFICATION: ARITHMETIC DISTRACTORS



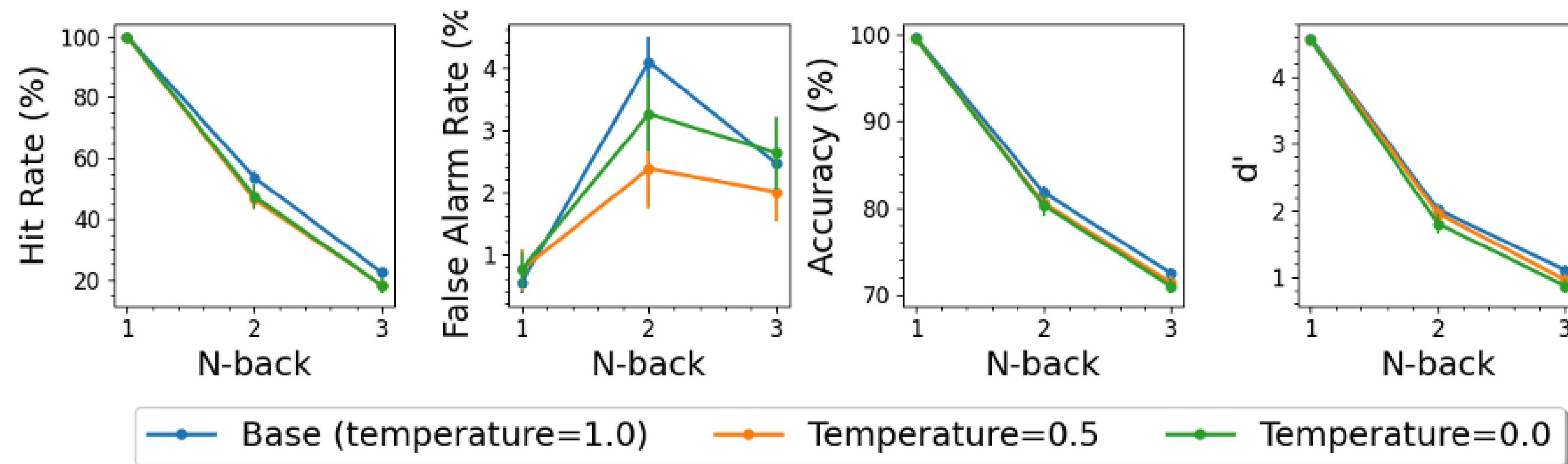
Interval k: Low (k=8), Medium (k=4), High (k=2).

RESULTS: IMPACT OF DISTRACTOR



Highlight: The stable performance at $k=8$, 4 followed by the sharp drop-off at $k=2$.

MINOR MODIFICATION & RESULTS: TEMPERATURE



Finding: d' stays nearly identical.

LESSONS AND RECOMMENDATIONS

- **Key Lesson:** Using n-back test to measure d' is a highly useful benchmark for quantifying "internal hardware" limits.
- **Recommendation 1: State Offloading.** Avoid using the context window as a primary database for active state.
- **Recommendation 2: Tool Calling, MCP & Agent Skills.** Use external protocols to manage memory-related tasks efficiently.

FINAL ASSESSMENT & SUMMARY

- **Section 1: What Can Be Reproduced?**
 - **The $n=3$ Decay Trend:** The performance decay at $n=3$ is a structural bottleneck.
- **Section 2: What Cannot Be Reproduced?**
 - **Absolute Metric Scores:** Modern models (DeepSeek-V3.2) provide a significantly higher baseline for $n=1$ and $n=2$, outperforming the original paper's benchmarks.
- **Section 3: Modification Key Finding**
 - **Distractor Sensitivity:** High-frequency distractors ($k=2$) degrade working memory state, proving that context "noise" is as critical as sequence length.



THANK YOU